

Isaac Mubiru

isaacmubiru99@gmail.com | linkedin.com/in/isaac-mubiru | github.com/zac-09 | Kampala, Uganda (Remote)

PROFESSIONAL SUMMARY

Backend engineer with 5+ years building **distributed, fault-tolerant systems** and running mission-critical workloads on the cloud. I architected an event-driven **microservices backend on Apache Kafka, Docker, and Kubernetes** for a national legal case-management platform, and built a live two-way **MongoDB↔Firestore sync pipeline on Google Pub/Sub** that kept a production store running through a zero-downtime migration of 20+ apps. Comfortable across **Python** and **Node.js/TypeScript**, NoSQL data stores, and Linux, with a bias for automation and reliability. Fully remote from **Kampala, Uganda (UTC+3, EMEA)**. Code: github.com/zac-09

CORE COMPETENCIES

Apache Kafka

Distributed Systems

Python

Docker & Kubernetes

NoSQL (MongoDB / Firestore)

Node.js / TypeScript

GCP / AWS

Linux

WORK EXPERIENCE

MTailor Sep 2022 – Present

Software Engineer · US Remote — custom-clothing e-commerce

- Shipped **Express Shipping** end-to-end in the checkout path, generating an additional **\$40 revenue per order**.
- Built a **3D product visualisation** feature using video overlay and ffmpeg, **increasing buyer conversion**.
- Led the end-to-end migration of **20+ production applications** from Parse/MongoDB to Firebase/**GCP** with **zero downtime**, reporting directly to the CTO.
- Implemented **real-time two-way sync** between MongoDB and Firestore using **Node.js** and Google **Pub/Sub**, keeping the live store running throughout the migration.
- Cut infrastructure cost by **\$5,000/month** by migrating all services off AWS; moved file storage from S3 to GCS with Python.

MTailor Mar 2022 – Jul 2022

Backend Contractor · US Remote

- Hired via Upwork for a complex backend migration; delivered a MongoDB→Firestore migration pipeline with complex processing logic and converted to a full-time role.
- Wrote engineer-facing documentation for the new Firebase SDKs (Node.js & Python) and trained the Ops team on dashboards and backups.

Dr Wealth Aug 2021 – Jan 2022

Software Engineer · Singapore Remote

- Optimised a PWA delivering **real-time** stock prices; built jobs querying **2M+ Firestore records** against Morningstar APIs.
- Extended backend APIs with Firebase Cloud Functions and Express.js; added responsive **React** + Tailwind pages.

Mind2matter Jan 2022 – Jul 2022

Software Engineer · US Remote

- Built backends for DeFi applications with **Node.js** and Web3; delivered **React** UIs from Figma designs under tight timelines.

CodeBits Jan 2020 – Jul 2021

Software Engineer / Team Lead · Uganda Remote

- Architected an event-driven **microservices backend** for FIDA Uganda's case-management platform using **Apache Kafka, Docker, and Kubernetes** — fault-tolerant messaging at national scale; led a team of 4 developers.
- Built a USSD legal-aid service backed by **Node.js** over **gRPC**, with **NATS Streaming** for inter-service events.
- Shipped a cross-platform mobile app with **React Native and Expo** (iOS & Android) with Firebase push notifications — one codebase, both stores.

PROJECTS

Real-time Pub/Sub Sync Pipeline

Live two-way MongoDB↔Firestore sync running under parallel production e-commerce traffic during a zero-downtime migration.

Node.js · Google Pub/Sub · MongoDB · GCP

LASPNET Mobile App — React Native + Expo

Cross-platform iOS/Android app shipped from a single codebase with Firebase push notifications.

React Native · Expo · Firebase

Microservices Backend — FIDA Uganda

Event-driven backend for a legal case-management platform serving NGOs at national scale.

Node.js · Apache Kafka · Docker · Kubernetes · gRPC

EDUCATION

B.Sc. Computer Engineering — Makerere University

2018 – 2022

Kampala, Uganda

LINKS

Open-source & project code — github.com/zac-09

SKILLS

Languages: JavaScript, TypeScript, Python, Go (intermediate), Rust (intermediate)

Frontend / Mobile: React, React Native, Expo, HTML, CSS, Tailwind **Backend:** Node.js, Express.js, REST APIs, gRPC, Microservices

Data: SQL, MongoDB, Firestore, Google Pub/Sub, Apache Kafka **Cloud / Infra:** GCP, Firebase, AWS, Docker, Kubernetes, Linux, Git

Spoken: English, Kiswahili